

Premier University, Chittagong.

Department of Computer Science and Engineering.



Final Project Report on

Google Play Store Applications Dataset Analysis.

Course: Big Data Analytics Laboratory.

Course Code: CSE 4346.

Submitted By:

Arnab Barua. (Id- 0222220005101091)

Submitted to: MD Tamim Hossain.

Lecturer, Department of CSE.

Premier University, Chittagong.

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Abstract

The increasing number of mobile applications has produced a large amount of user-generated data, making efficient data processing an important challenge. Traditional data processing methods become less effective when datasets grow significantly in size. Apache Hadoop provides a distributed environment that enables large-scale data storage and parallel processing.

This project analyzes a Google Play Store dataset stored in CSV format using Hadoop technologies. The dataset contains millions of application-related records, including application names, categories, ratings, review counts, installation numbers, pricing information, and content ratings. The complete workflow was implemented using HDFS for storage, Java MapReduce for distributed computation, and Apache Pig for high-level data analysis.

Three MapReduce programs and five Pig scripts were developed to perform different analytical tasks. These tasks include counting applications by content rating, calculating total reviews by category, identifying maximum installations, filtering free and paid applications, finding the most reviewed applications, counting category-wise applications, and identifying highly rated applications.

The project demonstrates how Hadoop components work together to process large datasets efficiently while producing meaningful analytical insights.

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1. Introduction

Big Data refers to datasets that are too large or complex to be processed efficiently using conventional data processing techniques. Modern digital platforms such as Google Play Store generate millions of application records that include ratings, downloads, reviews, and pricing information. Analyzing these records requires distributed computing techniques rather than traditional single-machine processing.

Apache Hadoop is one of the most widely used open-source frameworks for Big Data processing. It combines distributed storage through the Hadoop Distributed File System (HDFS) with parallel computation using the MapReduce programming model [1]. These technologies allow large datasets to be processed efficiently while improving scalability and fault tolerance.

In this laboratory project, the Google Play Store Applications Dataset was analyzed using Hadoop. Java MapReduce programs were implemented for computational tasks, while Apache Pig was used to simplify filtering, grouping, and sorting operations. The project provides practical experience with distributed data processing in a real-world environment.

2. Problem Statement

The Google Play Store contains applications belonging to thousands of different categories, each having different ratings, reviews, installation counts, and pricing models. Extracting useful insights from such a large dataset manually or using spreadsheet software is time-consuming and inefficient.

The main problem addressed in this project is how to efficiently process and analyze a large application dataset using Hadoop technologies. The project focuses on answering questions such as:

- Which content rating contains the highest number of applications?
- Which category has the highest total number of reviews?
- Which category contains applications with the maximum installations?
- How can free and paid applications be separated efficiently?
- Which applications receive the highest ratings?

These analytical tasks demonstrate the practical use of distributed computing for real-world datasets.

3. Objectives

The objectives of this project are:

- Configure a working Hadoop environment on Windows.
- Upload the Google Play Store dataset into HDFS.
- Process data using Java MapReduce.
- Analyze data using Apache Pig.
- Count applications by content rating.
- Calculate total reviews for each category.
- Identify maximum installations across categories.
- Filter free applications.
- Filter paid applications priced above five dollars.
- Find the most reviewed applications.
- Count applications for each category.
- Identify highly rated applications.

4. Dataset Description

This project uses the Google Play Store Dataset, collected from Kaggle [3]. The dataset is provided in CSV format with a file size of 676.46 MB and contains application-related information collected from the Google Play Store.

According to the dataset documentation, the dataset contains application information collected from the Google Play Store. The Data Explorer displays more than 2.17 million unique application names, 2.31 million unique App IDs, and 24 columns, making it suitable for large-scale data analysis.

The dataset includes important information such as application category, rating, review count, installation numbers, pricing details, developer information, and content rating. These attributes allow different types of analytical operations using Hadoop MapReduce and Apache Pig.

Before performing any analysis, the dataset was uploaded to the Hadoop Distributed File System (HDFS), where it was used as the input for all MapReduce programs and Pig scripts.

Dataset Information

Attribute	Description
Dataset Name	Google-Playstore.csv
Source	Kaggle
File Format	CSV
File Size	676.46 MB
Number of Columns	24
Unique App Names	2,177,946
Unique App IDs	2,312,944
Storage Platform	HDFS
Processing Tools	Hadoop MapReduce and Apache Pig

Major Attributes Used

Field	Purpose
App Name	Identifies each application
App ID	Unique package name
Category	Used for category-wise analysis
Rating	Used to identify highly rated apps
Rating Count	Used to calculate total reviews
Installs	Indicates application popularity
Minimum Installs	Used for installation analysis
Maximum Installs	Used for maximum install analysis
Free	Used to separate free and paid apps
Price	Used to filter paid applications
Content Rating	Used for audience-based analysis

These attributes were selected because they directly support the analytical tasks performed in this project, including counting applications by content rating, calculating total reviews by category, identifying highly rated applications, and analyzing installation trends.

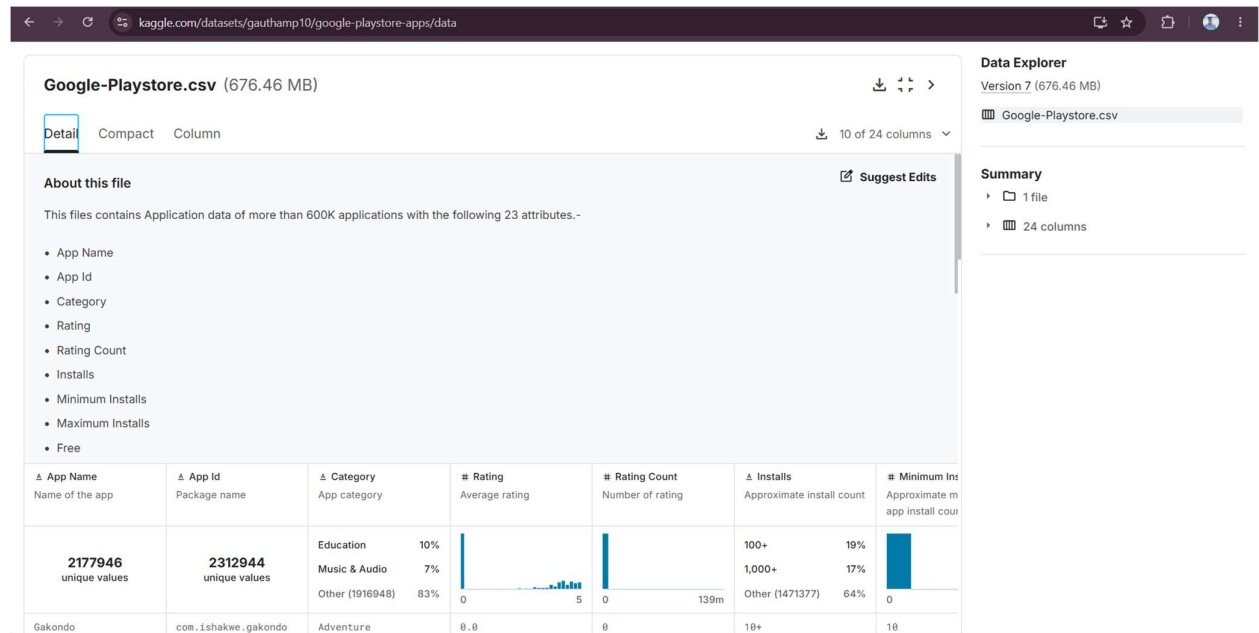


Figure 4.1: Google Play Store Dataset (Google-Playstore.csv)

5. Tools and Environment

Several software tools were used throughout the project.

Tool	Purpose
Windows 10	Operating System
Java JDK	Java compilation
Apache Hadoop	Big Data Framework
HDFS	Distributed Storage
MapReduce	Distributed Processing
Apache Pig	High-Level Data Analysis
Visual Studio Code	Code Development
Command Prompt	Command Execution
Browser	HDFS and YARN Monitoring

Hadoop Components Used

Hadoop Distributed File System (HDFS)

HDFS stores large datasets across distributed storage nodes while maintaining reliability through data replication [1].

MapReduce

MapReduce divides processing into two major phases:

- Mapper
- Reducer

This parallel execution significantly improves processing efficiency for large datasets [5].

Apache Pig

Apache Pig provides a high-level scripting language called Pig Latin that simplifies data analysis tasks compared to writing complete Java MapReduce programs [2].

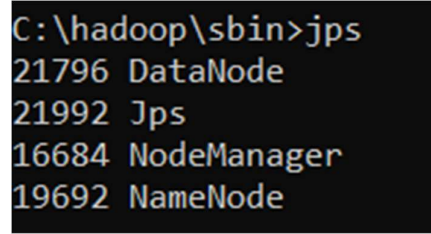
6. Hadoop System Setup

Before executing the analytical tasks, all Hadoop services were verified to ensure that the environment was functioning correctly.

The following Hadoop services were running successfully:

- NameNode
- DataNode
- ResourceManager
- NodeManager

The jps command was used to verify the running Java processes.



```
C:\hadoop\sbin>jps
21796 DataNode
21992 Jps
16684 NodeManager
19692 NameNode
```

Figure 6.1: Running Hadoop Services

7. HDFS Data Preparation

After confirming that Hadoop services were running correctly, the dataset was verified inside HDFS.

The root directory contained the project folder where the dataset and output directories were stored.

```
C:\hadoop\sbin>hdfs dfs -ls /  
Found 5 items  
drwxr-xr-x - User supergroup 0 2026-08-15 22:52 /BDA6  
drwxr-xr-x - User supergroup 0 2026-09-12 17:49 /PlayStoreProject  
drwx----- - User supergroup 0 2026-09-12 21:39 /tmp  
drwxr-xr-x - User supergroup 0 2026-08-21 22:01 /youtube  
drwxr-xr-x - User supergroup 0 2026-08-23 14:08 /youtube_output
```

Figure 7.1: HDFS Root Directory

The project directory contained separate folders for input and output data.

```
C:\hadoop\sbin>hdfs dfs -ls /PlayStoreProject  
Found 2 items  
drwxr-xr-x - User supergroup 0 2026-09-12 17:50 /PlayStoreProject/input  
drwxr-xr-x - User supergroup 0 2026-09-12 21:30 /PlayStoreProject/output
```

Figure 7.2: PlayStoreProject Directory

The input folder contained the uploaded Google Play Store dataset.

```
C:\hadoop\sbin>hdfs dfs -ls /PlayStoreProject/input  
Found 1 items  
-rw-r--r-- 1 User supergroup 58517326 2026-09-12 17:50 /PlayStoreProject/input/GooglePlay_200k.csv  
C:\hadoop\sbin>
```

Figure 7.3: Input Dataset Inside HDFS

Finally, the HDFS Explorer was used to verify the directory structure through the browser interface.

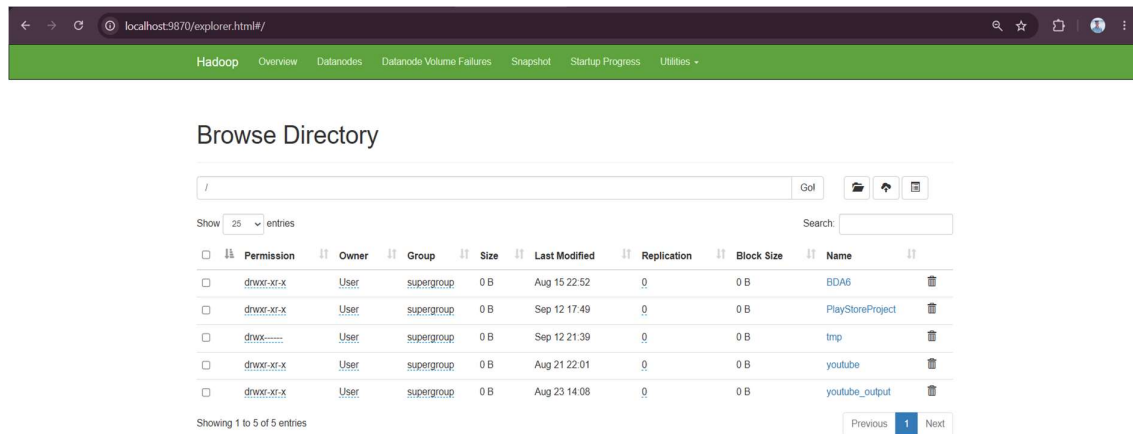


Figure 7.4: HDFS Explorer View

At this stage, the Hadoop environment was fully prepared for executing MapReduce programs and Apache Pig scripts.

8. MapReduce Analysis

Java MapReduce was used to perform three analytical tasks on the Google Play Store dataset stored in HDFS. Each task followed the same workflow:

1. The dataset was loaded from HDFS.
2. The Mapper extracted the required key-value pairs.
3. The Reducer aggregated or compared values.
4. The final output was stored back into HDFS.

8.1 Task 1: Content Rating Count

Objective

The objective of this task was to calculate the total number of applications available under each Content Rating category.

Examples include:

- Everyone
- Teen
- Mature 17+
- Everyone 10+

This helps understand which audience group contains the largest number of applications.

Working Procedure

The dataset was first read from HDFS.

The Mapper extracted the Content Rating field from each record and emitted:

(Content Rating, 1)

The Reducer received all values belonging to the same content rating and calculated the total count.

Driver Class

The Driver configured:

- Mapper
- Reducer
- Input Path
- Output Path

and submitted the job to Hadoop.

Mapper Logic

Each valid record generated one key-value pair.

Example:

Input	Output
Teen	(Teen,1)
Everyone	(Everyone,1)

Reducer Logic

The Reducer summed all values for each content rating.

Example:

Teen → 12000

Everyone → 85000

Execution Command

```
hadoop jar ContentRatingCount.jar ContentRatingCount  
/PlayStoreProject/input/GooglePlay_200k.csv  
/PlayStoreProject/output/MR1_ContentRatingCount
```

Output Verification

The output directory was verified successfully inside HDFS.

```
C:\hadoop\sbin>hdfs dfs -ls /PlayStoreProject/output/MR1_ContentRatingCount
Found 2 items
-rw-r--r--  1 User supergroup          0 2026-09-12 18:12 /PlayStoreProject/output/MR1_ContentRatingCount/_SUCCESS
-rw-r--r--  1 User supergroup        90 2026-09-12 18:12 /PlayStoreProject/output/MR1_ContentRatingCount/part-r-00000
```

Figure 8.1: MR1 Output Directory

The generated output file contained the final content rating counts.

```
C:\hadoop\sbin>hdfs dfs -cat /PlayStoreProject/output/MR1_ContentRatingCount/part-r-00000
Adults only 18+ 8
Everyone 174939
Everyone 10+ 2871
Mature 17+ 5092
Teen 17076
Unrated 14
```

Figure 8.2: MR1 Output

The same output was also verified through the HDFS browser.

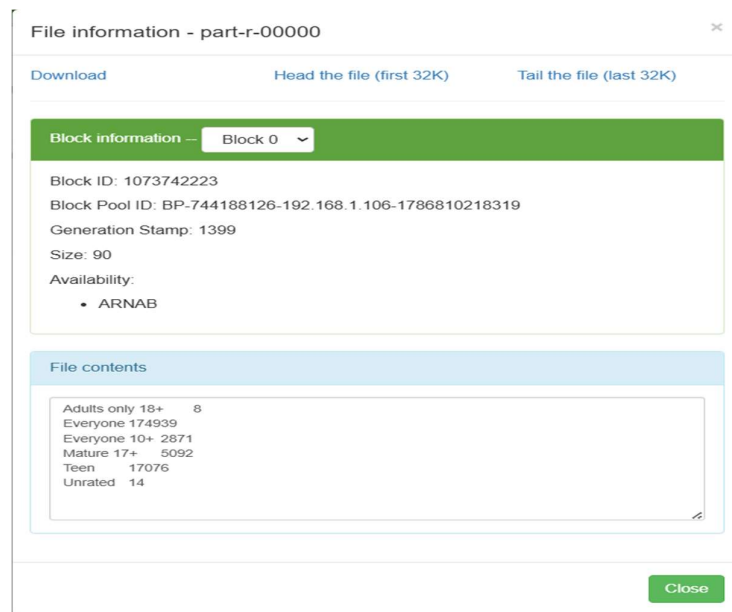


Figure 8.3: MR1 Output in HDFS Explorer

Observation

The output shows that the Everyone content rating contains the largest number of applications, indicating that most applications target a general audience.

8.2 Task 2: Total Reviews by Category

Objective

The purpose of this task was to calculate the total number of user reviews for each application category.

Instead of counting applications, this task measures overall user engagement within each category.

Working Procedure

The Mapper extracted:

- Category
- Rating Count

and emitted:

(Category, Review Count)

The Reducer added all review counts belonging to each category.

Driver Class

The Driver configured the job and stored the result in:

/PlayStoreProject/output/MR2_TotalReviewsByCategory

Mapper Logic

Example:

Category	Reviews
Action	500
Action	700

Output:

(Action,500)

(Action,700)

Reducer Logic

The Reducer produced:

Action → 1200

for each category.

Execution Command

```
hadoop jar TotalReviewsByCategory.jar TotalReviewsByCategory  
/PlayStoreProject/input/GooglePlay_200k.csv  
/PlayStoreProject/output/MR2_TotalReviewsByCategory
```

Output Verification

The output folder was successfully created.

```
C:\hadoop\sbin>hdfs dfs -ls /PlayStoreProject/output/MR2_TotalReviewsByCategory  
Found 2 items  
-rw-r--r--  1 User supergroup      0 2026-09-12 18:23 /PlayStoreProject/output/MR2_TotalReviewsByCategory/_SUCCESS  
-rw-r--r--  1 User supergroup    903 2026-09-12 18:23 /PlayStoreProject/output/MR2_TotalReviewsByCategory/part-r-000000
```

Figure 8.4: MR2 Output Directory

The output file displayed total review counts for every category.

```
C:\hadoop\sbin>hdfs dfs -cat /PlayStoreProject/output/MR2_TotalReviewsByCategory/part-r-000000 | more  
Action 90694134  
Adventure 13424969  
Arcade 13777383  
Art & Design 998283  
Auto & Vehicles 432303  
Beauty 279256  
Board 5651774  
Books & Reference 8913134
```

Figure 8.5: MR2 Output

The browser interface confirmed successful output generation.

```
C:\hadoop\sbin>hdfs dfs -cat /PlayStoreProject/output/MR2_TotalReviewsByCategory/part-r-000000 | more  
Action 90694134  
Adventure 13424969  
Arcade 13777383  
Art & Design 998283  
Auto & Vehicles 432303  
Beauty 279256  
Board 5651774  
Books & Reference 8913134
```

Figure 8.6: MR2 Output in HDFS Explorer

Observation

The results indicate that categories such as Social, Tools, and Entertainment received significantly higher review counts than many other categories, reflecting stronger user interaction.

8.3 Task 3: Maximum Installs by Category

Objective

The purpose of this task was to identify the maximum installation value found within each application category.

Unlike Task 2, this task compares installation numbers instead of adding them.

Working Procedure

The Mapper extracted:

- Category
- Maximum Installs

and emitted:

(Category, Maximum Installs)

The Reducer compared installation values and selected the highest value for each category.

Driver Class

The Driver managed the MapReduce job configuration and stored the output inside:

/PlayStoreProject/output/MR3_MaximumInstallsByCategory

Mapper Logic

Example:

Category	Installs
Social	100000
Social	500000

Output:

(Social,100000)

(Social,500000)

Reducer Logic

Instead of summation, the Reducer selected:

Social → 500000

which represents the highest installation value for that category.

Execution Command

```
hadoop jar MaximumInstallsByCategory.jar MaximumInstallsByCategory  
/PlayStoreProject/input/GooglePlay_200k.csv  
/PlayStoreProject/output/MR3_MaximumInstallsByCategory
```

Output Verification

The output folder was created successfully.

```
C:\hadoop\sbin>hdfs dfs -ls /PlayStoreProject/output/MR3_MaximumInstallsByCategory  
Found 2 items  
-rw-r--r-- 1 User supergroup 0 2026-09-12 18:46 /PlayStoreProject/output/MR3_MaximumInstallsByCategory/_SUCCESS  
-rw-r--r-- 1 User supergroup 968 2026-09-12 18:46 /PlayStoreProject/output/MR3_MaximumInstallsByCategory/part-r-00000
```

Figure 8.7: MR3 Output Directory

The generated file contained the highest installation value for every category.

```
C:\hadoop\sbin>hdfs dfs -cat /PlayStoreProject/output/MR3_MaximumInstallsByCategory/part-r-00000 | more  
Action 518325903  
Adventure 163379217  
Arcade 100228272  
Art & Design 34406443  
Auto & Vehicles 6852073  
Beauty 16085429  
Board 220484354  
Books & Reference 179730560  
Business 17717904  
Card 42034116
```

Figure 8.8: MR3 Output

The same output was verified through HDFS Explorer.

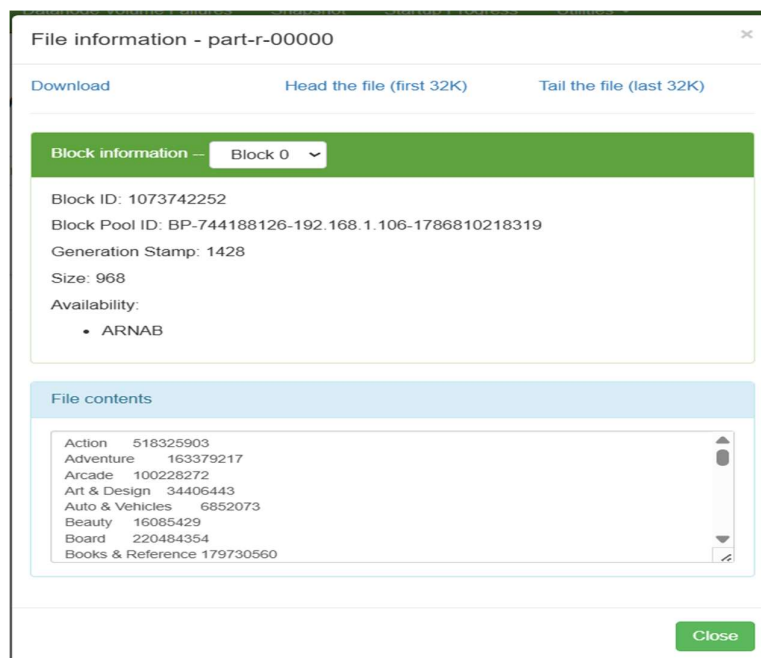


Figure 8.9: MR3 Output in HDFS Explorer

Observation

Several categories contained applications with extremely large installation numbers, including values reaching hundreds of millions of downloads, indicating widespread popularity among users.

MapReduce Summary

The three MapReduce programs performed different types of analytical operations while following the same distributed processing workflow.

Task	Output Produced
Content Rating Count	Total apps for each content rating
Total Reviews by Category	Sum of reviews for each category
Maximum Installs by Category	Highest install value in every category

Together, these tasks demonstrate the flexibility of MapReduce for performing different types of large-scale data analysis.

9. Apache Pig Analysis

Apache Pig was used to perform high-level data analysis on the Google Play Store dataset stored in HDFS. Instead of writing complete Java MapReduce programs for every analytical task, Pig Latin scripts were used to perform filtering, grouping, sorting, and counting operations with much simpler code.

The dataset was loaded from HDFS using `PigStorage(',')`, the header row was removed, and the required transformations were performed before storing the results back into HDFS.

9.1 Task 1: Free Applications

Objective

The objective of this task was to extract only the free applications from the Google Play Store dataset.

This analysis helps separate free applications from paid ones for further study.

Pig Script Logic

The script performed the following steps:

1. Load the dataset from HDFS.

2. Remove the header row.
3. Filter records where free = True.
4. Store the filtered records into HDFS.

The filtering operation was performed using Pig's FILTER statement.

Execution Command

`pig free_apps.pig`

Output Verification

The output directory was successfully created.

```
C:\hadoop\sbin>hdfs dfs -ls /PlayStoreProject/output/Pig1_FreeApps
Found 2 items
-rw-r--r-- 1 User supergroup 0 2026-09-12 18:59 /PlayStoreProject/output/Pig1_FreeApps/_SUCCESS
-rw-r--r-- 1 User supergroup 26485885 2026-09-12 18:59 /PlayStoreProject/output/Pig1_FreeApps/part-m-00000
```

Figure 9.1: Pig1 Output Directory

The output file contained only free applications.

```
C:\hadoop\sbin>hdfs dfs -cat /PlayStoreProject/output/Pig1_FreeApps/part-m-00000 | more
Gakondo,com.ishakwe.gakondo,Adventure,0.0.0,10+,10,15,True,0.0,USD,10M,7.1 and up,Jan Confident Ir-n-e NIYIZIBYOSE,https://beniyizibyoise.tk/#/,jean21181999@gmail.com,"Feb 26, 2020","Feb 26, 2020",Everyone,https://beniyizibyoise.tk/projects/,False,False
Vibook,com.doantiepvien.crm,Productivity,0.0.0,50+,50,50,True,0.0,USD,3.7M,4.0.3 and up,Cabin Crew,vnacrewit@gmail.com,"Aug 9, 2019","Aug 19, 2019",Everyone,https://www.vietnamairlines.com/vn/en/terms-and-conditions/privacy-policy,False,False
Smart City Trichy Public Service Vehicles 17UCS540,cst.stjoseph.ugl7ucs540,Communication,5.0.5,10+,10,19,True,0.0,USD,1.8M,4.0.3 and up,Climate Smart Tech2,http://www.climatesmarttech.com/,climatesmarttech2@gmail.com,"Sep 10, 2018","Oct 13, 2018",Everyone,True,False
GROM.me,com.horodyski.grower,Tools,0.0.0,100+,100,478,True,0.0,USD,6.2M,4.1 and up,Rafal Milek-Horodyski,http://www.horodyski.com.pl,rmlilekhorodyski@gmail.com,"Feb 21, 2020","Nov 12, 2018",Everyone,http://www.horodyski.com.pl,False,False
IMOCCL,com.imocci,ocial,0.0.0,50+,50,89,True,0.0,USD,46M,6.0 and up,Imocci GmbH,http://www.imocci.com,info@imocci.com,"Dec 24, 2018","Dec 20, 2019",Teen,https://www.imocci.com/wp-content/uploads/2018/08/Datenschutzerklrtrung_IMOCCL_22072018.pdf,False,True
The Everyday Calendar,com.mozaix.simonboard,Lifestyle,2.0.39,500+,500,702,True,0.0,USD,16M,5.0 and up,Mozaix LLC,elementuser03@gmail.com,"Jun 21, 2019","Jun 21, 2019",Everyone,https://www.freeprivacypolicy.com/privacy/view/978b22a2f432de423de81e4a91d571,False,False
whatsOpen,com.whatsopen.app,Communication,0.0.0,10+,10,18,True,0.0,USD,1.3M,4.4 and up,Yilver Molina Hurtatiz,http://yilvermolinah.com,yilver.mh1996@gmail.com,"Dec 07, 2018",Teen,http://elcafedelamanana.yilvermolinah.com/policy.html,False,False
Dodge The Cars!,com.MrScratchEnterprises.CarDogeGame,Racing,4.9.55,100+,100,329,True,0.0,USD,51M,4.4 and up,MrScratch,mithalaarush@gmail.com,"Jul 30, 2020","Jul 30, 2020",Everyone,https://docs.google.com/document/d/1HTTvenbFLk6ZrIRKzm8pYD1tFQ8hZMhFxrH8Qo_Xu/edit?usp=sharing,False,False
Parents.com.eqra.android.BSTS.parent,Maps & Navigation,0.0.0,100+,100,330,True,0.0,USD,2.7M,4.0 and up,Eqra Tech,http://www.hafizquran.com,hhiyassat@eqratech.com,"Jan 10, 2018","Jun 27, 2018",Everyone,https://docs.google.com/document/d/1fpof-YmfcQsZUQh0jZfj8983tCvc972aDukdFynjY/edit?usp=sharing,False,False
be.MOBILISED,com.hastobe.benobilised,Maps & Navigation,0.0.0,100+,100,274,True,0.0,USD,15M,6.0 and up,has.to.be.gmbh,http://www.has-to-be.com,support@has-to-be.com,"Feb 9, 2020","May 14, 2021",Everyone,https://has-to-be.com/de/ueber-uns/datenschutzrichtlinien/,False,False
OTENTIK Discovery FR,com.camneo.otentikdiscoveryOLFR,Travel & Local,3.4.5,500+,500,949,True,0.0,USD,2.9M,4.4 and up,Hotels Attitude,http://www.hotels-attitude.com,info@hotels-attitude.com,"Nov 28, 2016","Oct 30, 2019",Everyone,http://camneo.com/w7Wt699w8sdH2Fuy0eTQACM6mcm9Kf/CamneoPrivacy-v1.0.pdf,False,False
All in one shopping app,com.tkapplications.allinoneshoppingapp,Tools,5.0.6,50+,50,62,True,0.0,USD,2.0M,4.1 and up,TK Applications,,tkapplications001@gmail.com,"Apr 24, 2019","May 05, 2019",Everyone,False,False
Contemporary Love Wallpaper HD,com.felinddev.lowlpaper,Books & Reference,0.0.0,1+,1,2,True,0.0,XXX,11M,4.2 and up,Human Droid Apps,farukabdillah4@gmail.com,"Dec 26, 2020","Mar 23, 2021",Mature 17+,https://docs.google.com/document/d/1oKACClGgbyf11eHtV78HtIp9dKd0WbphZLlo8uGlc/edit?usp=sharing,True,False
Nepali Congress,pp.com.NepaliCongress.Social,0.0.0,100+,100,336,True,0.0,USD,24M,5.0 and up,Nitya Shree Developers,https://nepalicongress.org,info@nepalicongress.org,"May 31, 2021","May 31, 2021",Everyone,https://nepalicongress.org/page/privacy-policy,False,False
Bhavan's BHEL VIM,com.bhavans.vim,Medical,0.0.0,10+,10,31,True,0.0,USD,1.4M,4.0.3 and up,Bhavans, BHEL, Hyderabad,http://www.bhavansbhelhyd.com,cheboulou.uma@gmail.com,"Dec 25, 2017","Dec 26, 2017",Everyone,Sticky Ball,com.mitochondriastudios.stickyball,Arcade,4.6.11,100+,100,152,True,0.0,USD,25M,4.1 and up,Mitochondria Studios,family@mitochondriastudios.com,"Mar 5, 2018","Mar 26, 2018",Everyone,https://mitoch--
-- More -- Lines: 1
```

Figure 9.2: Free Applications Output

The Pig script used for this task is shown below.

```
apps = LOAD '/PlayStoreProject/input/GooglePlay_200k.csv'
USING PigStorage(',')
AS (
  app:chararray,
  appid:chararray,
  category:chararray,
  rating:float,
  ratingcount:int,
  installs:chararray,
  mininstalls:long,
  maxinstalls:long,
  free:chararray,
  price:float,
  currency:chararray,
  size:chararray,
  android:chararray,
  developer:chararray,
  website:chararray,
  email:chararray,
  released:chararray,
  updated:chararray,
  content:chararray,
  privacy:chararray,
  ads:chararray,
  inapp:chararray,
  editor:chararray,
  scraped:chararray
);

-- Header remove
data = FILTER apps BY NOT (app == 'App Name');

-- Free apps
freeapps = FILTER data BY free == 'True';

STORE freeapps INTO '/PlayStoreProject/output/Pig1'
USING PigStorage(',');
```

Figure 9.3: Pig1 Script

Observation

Most applications in the dataset belong to the free application category, which reflects the common business model of mobile applications where revenue is generated through advertisements or in-app purchases instead of direct pricing.

9.2 Task 2: Paid Applications with Price Above \$5

Objective

The purpose of this task was to identify paid applications whose price is greater than \$5.

This filtering operation helps analyze premium applications within the dataset.

Pig Script Logic

The script:

- Loaded the dataset
- Removed the header
- Filtered applications where:

price > 5

- Stored the resulting records into HDFS.

Execution Command

pig paid_apps.pig

Output Verification

The output folder was successfully created.

```
C:\hadoop\sbin>hdfs dfs -ls /PlayStoreProject/output/Pig2_PaidAppsPriceAbove5
Found 2 items
-rw-r--r-- 1 User supergroup 0 2026-09-12 19:47 /PlayStoreProject/output/Pig2_PaidAppsPriceAbove5/_SUCCESS
-rw-r--r-- 1 User supergroup 125235 2026-09-12 19:47 /PlayStoreProject/output/Pig2_PaidAppsPriceAbove5/part-m-00000
```

Figure 9.4: Pig2 Output Directory

The generated output contained paid applications priced above five dollars.

```
C:\hadoop\sbin>hdfs dfs -cat /PlayStoreProject/output/Pig2_PaidAppsPriceAbove5/part-m-00000 | more
VB CODE PRIME,appinventor.ai.mc207742.K8.CODE.PRIME,Tools,0.0,0.10+,10.30,False,24.99,USD,4.50,2.1 and up,Marcelo Santos Carvalho Nogueira,mc207742@gmail.com,"Aug 8, 2019","Aug 09, 2019",Everyone,,False,False
Clay Shooting Trap Shoot,com.cseg.tshoot,Sports,4.2,6.100+,100,254,False,5.49,USD,100,7.0 and up,Clay Shooting Apps,http://www.clayshootingindgym.com,info@clayshootingindgym.com,"Feb 14, 2014","Apr 15, 2020",Everyone,http://www.clayshootingindgym.com/privacy,False,False
Amazing Wheels Pro (No Ads),net.xmano.android.amazingwheel.tv,Tools,4.0,6.100+,100,328,False,5.49,USD,4.70,5.0 and up,Banana Studio,https://xmano.net,support@xmano.net,"Jan 5, 2020","Jan 20, 2020",Everyone,https://xmano.net/privacy/privacy_policy.html,False,False
Fire Installer Pro Donate,com.dlka.fireinstaller2,Tools,3.6,16.500+,500,642,False,6.49,USD,Varies with device,Varies with device,DLX Appentwicklung (Kulisch-IT),https://github.com/TheLandOILYdave/fireinstaller,TheLandOILYdave@googlemail.com,"Jan 17, 2015","Dec 15, 2015",Everyone,,False,True
Social Media Marketing Pro,in.softecks.socialmediamarketing.pro,Education,0.0,0.0+,5.5,False,5.29,USD,7.70,4.4 and up,Inteltech,http://www.softecks.in,contact@softecks.in,"Dec 21, 2019","Dec 21, 2019",Everyone,http://inteltech.in/privacy_policy.html,False,False
Metal Calculator PRO,com.dahiler.metalcalculator,Business,4.9,11.100+,100,220,False,9.99,USD,3.00,4.0.3 and up,Dahiler,http://www.dahiler.com,play@dahiler.com,"Mar 31, 2016","Jul 06, 2017",Everyone,https://www.cadcamsektor.com/privacy_policy.html,False,False
Montana Hunting Access 2020,com.emountainworks.android.huntingaccess,Sports,4.7,31.500+,500,815,False,9.99,USD,5330,5.0 and up,Mountainworks Software Inc,http://www.emountainworks.com,feedback@mountainworks.com,"Sep 10, 2015","Apr 01, 2021",Everyone,http://emountainworks.com/privacypolicy.aspx,False,False
Malayalam Dictionary Pro - Offline and Bilingual,com.orchid.malayalamdictionarypro,Education,4.3,9.100+,100,104,False,9.99,USD,Varies with device,Varies with device,Orchid Technologies,http://orchidtechnologies.in/android-english-malayalam-dictionary/contact@orchidtechnologies.in,"Feb 10, 2013","Mar 11, 2021",Everyone,,False,False
The Birds of Peru,com.in.docu.pebirds,Books & Reference,0.0,0.1+,1.1,False,8.99,USD,740,4.4 and up,Ursula Schmigelski,http://www.in-docu.com,ursula.schmigelski@gmail.com,"Jun 21, 2020","Jun 21, 2020",Everyone,http://www.in-docu.net/privacy_policy.html,False,False
Sociology Quiz Prep Pro,com.prepinteractive.Sociologypro,Education,0.0,0.0+,0.0,False,28.99,USD,6.30,5.0 and up,Prep Interactive,https://sites.google.com/view/prepinteractive,interactiveprep@gmail.com,"Jul 29, 2019","Jul 29, 2019",Everyone,https://sites.google.com/view/prepinteractive,False,False
iLanXit: 10-402-001-10-4 Pro,kh.taxitestpaid,Tools,3.8,6.100+,100,163,False,7.49,USD,310,4.2 and up,Solusworks Limited,admin@solusworks.com,"Jan 18, 2019","Dec 13, 2019",Everyone,https://s3-ap-northeast-1.amazonaws.com/hkgovjobs/privacy_policy.html,False,False
Paladin - Turn Based Fantasy Combat,com.tarodev.Paladin.Strategy,4.4,32.100+,100,395,False,7.99,USD,290,4.1 and up,Tarodev,,matt@tarodev.com,"Nov 5, 2019","Jan 14, 2020",Everyone,10+,https://matthew-j-spencer.github.io/Policies/Paladin.html,False,True
Oxford Handbook Clin Pharma 2e,net.medhand.ohcp2,Medical,0.0,0.10+,10,43,False,48.95,USD,5.40,4.0 and up,Indextra AB,http://www.medhand.com,android@medhand.net,"Aug 26, 2013","Jan 27, 2017",Everyone,https://bookstore.medhand.com/info/privacy,False,False
Bracketing for GoPro Hero5,com.toolsforgopro.bracketing.Photography,0.0,0.10+,10,21,False,5.99,USD,430,5.0 and up,It-Apps,https://www.toolsforgopro.com/bracketing,office@mobwheel.com,"Nov 16, 2020","May 17, 2021",Everyone,https://www.toolsforgopro.com/privacy,False,False
Red Centre - Let'sGoToMU.com,com.darkduck.letsgettoMU.uluru.Travel & Local,0.0,0.10+,10,36,False,5.31,USD,120,4.1 and up,Let's Go To Australia!,thenomadiceplorers@gmail.com,"Apr 11, 2016","Apr 12, 2016",Everyone,,False,False
Legend Pro - Intro Maker,com.animate.legendpro,Entertainment,3.1,24.500+,500,739,False,6.99,USD,400,4.3 and up,Dream Box Inc,http://visioningenioustech.com/,lavan.cct@gmail.com,"Apr 29, 2019","Jun 14, 2019",Everyone,http://visioningenioustech.com/,False,False
Hilfsterbuch Griechisch PREMIUM von PONS,com.sloved.noreg.pons.german_greek,Books & Reference,4.2,26.500+,500,623,False,54.99,USD,410,4.4 and up,PONS,http://www.pons.de,software-support@pons.de,"Aug 2, 2010","Mar 18, 2020",Teen,http://d.pons.com/shop/datenschutz.en.do,False,False
Forex Trading Signals and Alerts Daily App Premium,com.lastbuildit.forexsignals,Finance,4.2,125.500+,500,707,False,11.99,USD,110,4.1 and up,lastbuildIT,lastbuildit@gmail.com,"Nov 2, 2019","Jun 27, 2020",Teen,http://forexsolution.xyz/privacy.txt,False,False
Urdu English Keyboard with Photo Background Pro,com.urdu_keyboard.pro,Productivity,4.4,7.100+,100,159,False,5.99,USD,-- More --
```

Figure 9.5: Paid Applications Output

The corresponding Pig script is shown below.

```
apps = LOAD '/PlayStoreProject/input/GooglePlay_200k.csv'
USING PigStorage(',')
AS (
  app:chararray,
  appid:chararray,
  category:chararray,
  rating:float,
  ratingcount:int,
  installs:chararray,
  mininstalls:long,
  maxinstalls:long,
  free:chararray,
  price:float,
  currency:chararray,
  size:chararray,
  android:chararray,
  developer:chararray,
  website:chararray,
  email:chararray,
  released:chararray,
  updated:chararray,
  content:chararray,
  privacy:chararray,
  ads:chararray,
  inapp:chararray,
  editor:chararray,
  scraped:chararray
);

data = FILTER apps BY NOT (app == 'App Name');

paid = FILTER data BY free == 'False' AND price > 5;

STORE paid INTO '/PlayStoreProject/output/Pig2_PaidAppsPriceAbove5'
USING PigStorage(',');
```

Figure 9.6: Pig2 Script

Observation

The filtered results show that only a relatively small portion of applications fall into the higher-price category, indicating that expensive paid applications are less common than free applications.

9.3 Task 3: Top 10 Most Reviewed Applications

Objective

This task identifies the top ten applications with the highest review counts.

Applications with a large number of reviews generally indicate higher user engagement.

Pig Script Logic

The script performed these operations:

1. Load dataset
2. Remove header

3. Order records by review count in descending order
4. Select the first ten records
5. Store the output

The ORDER operation was used to rank applications.

Execution Command

`pig top_reviews.pig`

Output Verification

The output directory was successfully created.

```
C:\hadoop\sbin>hdfs dfs -ls /PlayStoreProject/output/Pig3_Top10MostReviewedApps
Found 2 items
-rw-r--r-- 1 User supergroup 0 2026-09-12 20:18 /PlayStoreProject/output/Pig3_Top10MostReviewedApps/_SUCCESS
-rw-r--r-- 1 User supergroup 2777 2026-09-12 20:18 /PlayStoreProject/output/Pig3_Top10MostReviewedApps/part-r-00000
```

Figure 9.7: Pig3 Output Directory

The output file contained the top ten most reviewed applications.

```
C:\hadoop\sbin>hdfs dfs -cat /PlayStoreProject/output/Pig3_Top10MostReviewedApps/part-r-00000 | more
PUBG MOBILE - Traverse,com.tencent.ig,Action,4.3,37479811,"500,0,500000000,5.05107228,True,0,USD,Varies with device,Varies with device,PROXIMA BETA,,PUBGMOBILE_C$@tencentgames.com,"Mar 19, 2018","May 10, 2021",Teen,http://pubgmobile.proximabeta.com/privacy.html
TikTok.com.chillaxapp.musically.Social,4.4,36446301,"1,0,0,000",1.069,1645011582,True,0,USD,69M,4.4 and up,TikTok Pte. Ltd.,https://www.tiktok.com/feedback@tiktok.com,"Jul 9, 2019","Jun 15, 2021",Teen
Shadow Fight 2,com.nekki.shadowfight,Action,4.6,14291156,"100,0,100000000,2.981371048,True,0,USD,140M,4.4 and up,neki - Action and Fighting Games,http://support.shadowfight2.com/support@neki.mall.helpshift.com,"Jul 22, 2014","May 19, 2021",Everyone 10+,http://neki.r
u/privacy.php
Among Us,com.innersloth.spacemafia,Action,3.5,12022649,"100,0,100000000,4.05421128,True,0,USD,110M,6.8 and up,Innersloth LLC,https://www.innersloth.com/support@innersloth.com,"Jun 15, 2018","Jun 14, 2021",Everyone 10+,https://www.innersloth.com/privacy.php
Netflix.com.netflix.mediaclient.Entertainment,4.4,11634633,"1,0,0,000",1.069,1497250178,True,0,USD,Varies with device,Varies with device,Netflix, Inc.,http://www.netflix.com/playstore@netflix.com,"May 5, 2011","Jun 14, 2021"
Pinterest.com.pinterest.Lifestyle,4.6,8017109,"500,0,500000000,6.15144963,True,0,USD,Varies with device,Varies with device,Pinterest,https://help.pinterest.com/help@pinterest.com,"Aug 14, 2012","Jun 14, 2021",Teen,https://policy.pinterest.com/privacy-policy
Gangstar Vegas: World of Crime,com.gameloft.android.ANMP.GloftGOW,Action,4.3,6278522,"100,0,100000000,1.092364083,True,0,USD,44M,4.4 and up,Gameloft SE,http://www.gameloft.com/android.support@gameloft.com,"Jun 19, 2013","May 25, 2021",Mature 17+,http://www.gameloft.c
om/privacy-notice/
Criminal Case,com.prettysimple.criminalcase.android,Adventure,4.5,4275556,"100,0,100000000,1.633792168,True,0,USD,Varies with device,4.1 and up,Pretty Simple,https://www.prettysimplegames.com/help@prettysimplegames.com,"Apr 15, 2015","Dec 03, 2020",Teen,http://www.pret
tysimplegames.com/about/privacy-policy/
YouCam Makeup - Selfie Editor & Magic Makeup Cam,com.cyberlink.youcammakeup,Photography,4.5,4023086,"100,0,100000000,2.861294728,True,0,USD,Varies with device,Varies with device,Perfect Mobile Corp.,https://www.perfectcorp.com/consumer/YouCamMakeup_android@perfectcor
p.com,"Aug 7, 2014","Jun 06, 2021",Everyone,https://www.perfectcorp.com/stat/BeautyCircle/en/privacy.html?utm_source=mkduta_medium-androidduta_campaign-policypage
Toon Blast,net.peakgames.toonblast,Puzzle,4.6,3237085,"50,0,50000000,7.707246467,True,0,USD,140M,4.4 and up,Peak,https://peakgames.helpshift.com/a/toon-blast/,support@peak.com,"Aug 25, 2017","Jun 11, 2021",Everyone,https://peak.com/privacy
```

Figure 9.8: Top Reviewed Applications Output

The Pig script is shown below.

```
apps = LOAD '/PlayStoreProject/input/GooglePlay_200k.csv'
USING PigStorage(',')
AS (
  app:chararray,
  appid:chararray,
  category:chararray,
  rating:float,
  ratingcount:int,
  installs:chararray,
  mininstalls:long,
  maxinstalls:long,
  free:chararray,
  price:float,
  currency:chararray,
  size:chararray,
  android:chararray,
  developer:chararray,
  website:chararray,
  email:chararray,
  released:chararray,
  updated:chararray,
  content:chararray,
  privacy:chararray,
  ads:chararray,
  inapp:chararray,
  editor:chararray,
  scraped:chararray
);

data = FILTER apps BY NOT (app == 'App Name');

ordered = ORDER data BY ratingcount DESC;

top10 = LIMIT ordered 10;

STORE top10 INTO '/PlayStoreProject/output/Pig3_Top10MostReviewedApps'
USING PigStorage(',');
```

Figure 9.9: Pig3 Script

Observation

Applications such as PUBG Mobile, TikTok, and Netflix appeared among the highest reviewed applications, indicating their widespread popularity and active user communities.

9.4 Task 4: Category-wise Application Count

Objective

The objective of this task was to calculate the number of applications available in each category.

This provides an overview of category distribution within the dataset.

Pig Script Logic

The script performed:

- Load dataset
- Remove header
- Group records by category
- Count applications
- Store the result

The GROUP and COUNT operations were used together.

Execution Command

pig category_count.pig

Output Verification

The output directory was successfully created.

```
C:\hadoop\sbin>hdfs dfs -ls /PlayStoreProject/output/Pig4_CategoryWiseAppCount
Found 2 items
-rw-r--r--  1 User supergroup          0 2026-09-12 20:46 /PlayStoreProject/output/Pig4_CategoryWiseAppCount/_SUCCESS
-rw-r--r--  1 User supergroup 139408 2026-09-12 20:46 /PlayStoreProject/output/Pig4_CategoryWiseAppCount/part-r-00000
```

Figure 9.10: Pig4 Output Directory

The output file displayed category-wise application counts.

```
C:\hadoop\sbin>hdfs dfs -cat /PlayStoreProject/output/Pig4_CategoryWiseAppCount/part-r-00000 | more
Education,20060
Music & Audio,13350
Business,12219
Tools,12092
Entertainment,11452
Lifestyle,10068
Books & Reference,9671
Personalization,7641
Health & Fitness,7034
```

Figure 9.11: Category Count Output

The script used for this analysis is shown below.

```
app:chararray,
appid:chararray,
category:chararray,
rating:float,
ratingcount:int,
installs:chararray,
mininstalls:long,
maxinstalls:long,
free:chararray,
price:float,
currency:chararray,
size:chararray,
android:chararray,
developer:chararray,
website:chararray,
email:chararray,
released:chararray,
updated:chararray,
content:chararray,
privacy:chararray,
ads:chararray,
inapp:chararray,
editor:chararray,
scraped:chararray
);

-- Remove header
data = FILTER apps BY NOT (app == 'App Name');

-- Group by category
grp = GROUP data BY category;

-- Count apps in each category
result = FOREACH grp GENERATE group AS category, COUNT(data) AS total_apps;

-- Sort by highest count
ordered = ORDER result BY total_apps DESC;

STORE ordered INTO '/PlayStoreProject/output/Pig4_CategoryWiseAppCount'
USING PigStorage(',');
```

Figure 9.12: Pig4 Script

Observation

The results show that some categories contain significantly more applications than others, demonstrating an uneven distribution of applications across different categories.

9.5 Task 5: Highly Rated Applications

Objective

The purpose of this task was to identify applications with a rating of 4.8 or higher.

This analysis highlights applications that receive consistently high user satisfaction.

Pig Script Logic

The script executed the following steps:

1. Load dataset.
2. Remove the header row.
3. Filter applications where:

rating $\geq$ 4.8

4. Store the filtered records into HDFS.

Unlike the previous tasks, this analysis focuses on quality rather than popularity.

Execution Command

pig topRated.pig

Output Verification

The output folder was successfully created.

```
C:\hadoop\sbin>hdfs dfs -ls /PlayStoreProject/output/Pig5_TopRatedApps
Found 2 items
-rw-r--r--  1 User supergroup      0 2026-09-12 21:31 /PlayStoreProject/output/Pig5_TopRatedApps/_SUCCESS
-rw-r--r--  1 User supergroup 719860 2026-09-12 21:31 /PlayStoreProject/output/Pig5_TopRatedApps/part-m-00000
```

Figure 9.13: Pig5 Output Directory

The output file contained highly rated applications.

```
C:\hadoop\sbin>hdfs dfs -cat /PlayStoreProject/output/Pig5_TopRatedApps/part-m-00000 | more
Smart City Trichy Public Service Vehicles 17UCS548,Communication,5.0,5
Dodge The Cars!,Racing,4.9,55
All in one shopping app,Tools,5.0,6
BuradaOto,Shopping,4.9,102
Niagara Falls Wallpapers,Personalization,5.0,11
Extrude Balance,Arcade,5.0,8
Coffin Move,Casual,4.9,8
Triple Point Academy,Education,5.0,5
```

Figure 9.14: Highly Rated Applications Output

The Pig script used for this task is shown below.

```
apps = LOAD '/PlayStoreProject/input/GooglePlay_200k.csv'
USING PigStorage(',')
AS (
  app:chararray,
  appid:chararray,
  category:chararray,
  rating:float,
  ratingcount:int,
  installs:chararray,
  mininstalls:long,
  maxinstalls:long,
  free:chararray,
  price:float,
  currency:chararray,
  size:chararray,
  android:chararray,
  developer:chararray,
  website:chararray,
  email:chararray,
  released:chararray,
  updated:chararray,
  content:chararray,
  privacy:chararray,
  ads:chararray,
  inapp:chararray,
  editor:chararray,
  scraped:chararray
);

data = FILTER apps BY NOT (app == 'App Name');

toprated = FILTER data BY rating >= 4.8;

result = FOREACH toprated GENERATE
  app,
  category,
  rating,
  ratingcount;

STORE result INTO '/PlayStoreProject/output/Pig5_TopRatedApps'
USING PigStorage(',')|
```

Figure 9.15: Pig5 Script

Observation

The filtered results indicate that applications with ratings of 4.8 and above are distributed across multiple categories rather than being concentrated in a single category.

Apache Pig Summary

The five Pig scripts demonstrate how Pig Latin simplifies common data analysis tasks that would otherwise require significantly more Java code.

Task	Main Operation	Output
Pig1	FILTER	Free apps
Pig2	FILTER	Paid apps above \$5
Pig3	ORDER + LIMIT	Top 10 reviewed apps
Pig4	GROUP + COUNT	Category counts
Pig5	FILTER	Apps rated ≥ 4.8

Compared to Java MapReduce, Pig required fewer lines of code while producing the same analytical outputs for filtering and grouping tasks.

Transition to Results

The MapReduce and Pig analyses together produced multiple outputs that reveal different aspects of the Google Play Store dataset. The next section compares these results and discusses the overall findings of the project.

10. Results and Discussion

After completing all MapReduce and Apache Pig tasks, the Google Play Store dataset was successfully processed and analyzed using the Hadoop ecosystem. The generated outputs provide insights into application distribution, popularity, pricing patterns, and user engagement.

The project demonstrates that Hadoop can efficiently process a large dataset by dividing the workload into distributed operations. MapReduce handled aggregation and comparison tasks effectively, while Pig simplified filtering and grouping operations with fewer lines of code.

Overall Output Structure

The final HDFS output directory contained all completed MapReduce and Pig outputs.

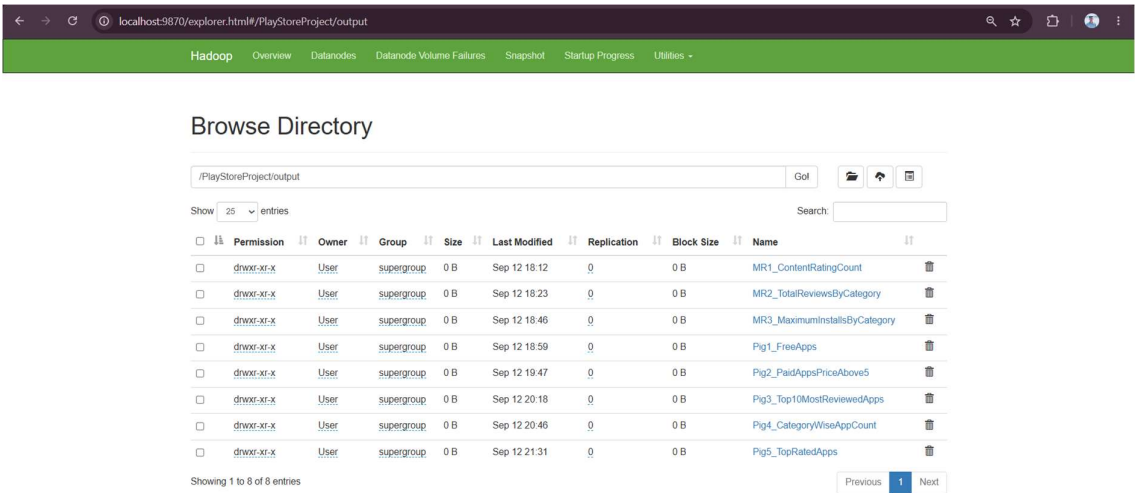


Figure 10.1: Final Output Directory in HDFS

Key Findings

The completed analyses revealed several interesting observations from the dataset.

Content Rating Distribution

The Content Rating Count task showed that the Everyone category contains the highest number of applications, indicating that most developers target a general audience.

User Engagement

The Total Reviews by Category task revealed that categories such as Social, Tools, and Entertainment received significantly higher review counts than many other categories, reflecting stronger user interaction.

Installation Trends

The Maximum Installs by Category analysis identified categories containing applications with hundreds of millions of installations, demonstrating the widespread popularity of certain mobile applications.

Pricing Pattern

The Pig filtering tasks showed that free applications dominate the dataset, while paid applications priced above \ \$5 represent only a small portion of the available applications.

Popular Applications

The Top Reviewed Applications analysis highlighted globally popular applications such as PUBG Mobile, TikTok, Netflix, and Pinterest, confirming that applications with massive user communities also receive extremely high review counts.

Highly Rated Applications

The highly rated applications analysis showed that applications with ratings above 4.8 are distributed across multiple categories rather than being concentrated in only one category.

11. Comparison Between MapReduce and Apache Pig

Both MapReduce and Pig were used to process the same dataset, but each approach has its own strengths.

Feature	MapReduce	Apache Pig
Programming	Java	Pig Latin
Complexity	Higher	Lower
Code Length	Longer	Shorter
Development Speed	Slower	Faster
Aggregation	Excellent	Excellent
Filtering	Possible	Simpler
Sorting	Manual implementation	Built-in

MapReduce provided greater flexibility for implementing custom processing logic, while Pig significantly reduced development time for common analytical operations.

For educational purposes, using both technologies together provides a better understanding of different approaches to Big Data processing.

12. Challenges Faced During Implementation

Several practical challenges were encountered while completing this project on a Windows-based Hadoop environment.

12.1 Hadoop Services Stopped Unexpectedly

At one stage, the administrator Command Prompt was accidentally closed, causing Hadoop services to stop. The issue was resolved by restarting the required services using the Hadoop startup scripts.

12.2 MapReduce Job Stuck at 0%

One MapReduce job remained at 0% completion because the Hadoop services were not running correctly. Restarting NameNode, DataNode, ResourceManager, and NodeManager solved the problem.

12.3 Pig Script Path Error

Initially, Pig scripts produced path-related errors when Windows file paths were used directly. The scripts were executed successfully after using the correct execution method from the project directory.

12.4 Job History Server Warning

During Pig execution, repeated retry messages appeared while connecting to port 10020. Although these messages continued after job completion, the outputs were successfully generated in HDFS. This behavior is common in Windows-based Hadoop installations when the Job History Server is not fully configured.

13. Advantages of the Project

The project demonstrates several practical advantages of using Hadoop technologies.

- Efficient processing of a large dataset.

- Distributed storage using HDFS.
- Parallel computation through MapReduce.
- Simplified data analysis using Apache Pig.
- Real-world experience with Hadoop ecosystem components.
- Practical understanding of distributed computing concepts.

14. Limitations

Although the project was successfully completed, several limitations remain.

- The implementation uses a single-node Hadoop cluster rather than a true multi-node cluster.
- Windows-based Hadoop configuration requires additional manual setup.
- CSV formatting issues may occur with complex text fields.
- The dataset is static and does not represent real-time application updates.

15. Future Work

The project can be extended in several ways.

- Implement Apache Spark for faster processing.
- Analyze larger datasets containing millions of records.
- Build a multi-node Hadoop cluster.
- Create interactive dashboards for visualization.
- Perform machine learning analysis using Spark MLlib.

These improvements would make the system more suitable for production-scale Big Data analysis.

16. Conclusion

This project successfully demonstrated a complete Big Data processing workflow using Apache Hadoop, Java MapReduce, and Apache Pig.

The Google Play Store dataset was stored in HDFS, processed through distributed computation, and analyzed using both low-level and high-level data processing techniques. Three MapReduce programs and five Pig scripts were

successfully implemented to answer different analytical questions related to application categories, ratings, reviews, installations, and pricing.

The project also provided practical experience in configuring Hadoop services, executing distributed jobs, troubleshooting execution issues, and managing HDFS outputs. Overall, the project illustrates how Hadoop technologies can efficiently process large datasets while producing meaningful insights from real-world application data.

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